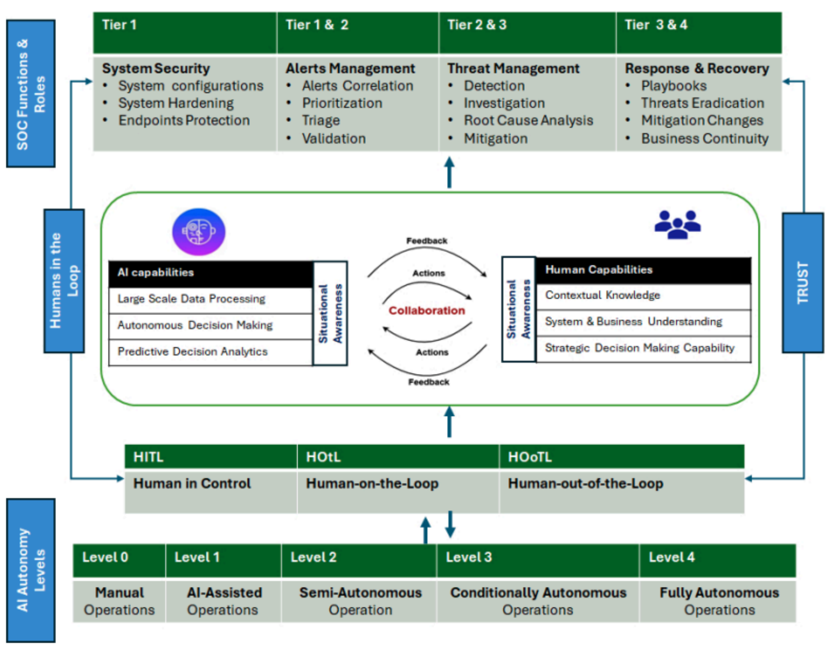
Reducing SOC False Alarms through a Human-AI Collaboration Model

Background

The high rate of false positives in the Security Operations Center (SOC) is often caused by the *Better Safe Than Sorry* philosophy, which prioritizes recall to ensure no threats are missed. This increases system sensitivity but also triggers excessive alerts, leading to burnout for the SOC team. Furthermore, the dynamic nature of cyber threats makes it difficult to manually differentiate between safe and malicious activity.

Project Task

Develop a Human-AI Collaboration SOC system that reduces false alarms without compromising detection accuracy.



Project Specifications

1. **Team Composition:** 1 group consisting of 6-8 people.
2. **Architecture:** The system configuration integrates a Wazuh Manager unit, a SOAR platform, and several Wazuh Agents, all operated on the Azure free-tier student server infrastructure.

3. **Scenario:** The scenario includes DDoS, Malware, and Social Engineering attacks (free case selection).
4. **Analysis:** Independently define the criteria for false alarms based on data from Wazuh.
5. **AI Integration:** Integrate an AI model into the Wazuh architecture to reduce false alarms. The development of the AI model must be done independently, and the use of third-party API services is not permitted.
6. **SOAR:** Use SOAR to prove that the architecture is still capable of responding effectively to attacks.

Deliverables

1. **Implementation:** GitHub Repository (source code)
2. **Documentation:** Final report including:
    - Details of the implemented architecture
    - Clear architectural images/diagrams
    - Explanation of the AI used
    - Benchmark metrics used
    - Analysis of the results obtained